Systems and Methods Using Density Field Dynamics for Information Encoding, Transmission, and Computing Applications

Gary Alcock

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Cross-References to Related Applications

This application claims the benefit of U.S. Provisional Patent Application No. 63/865,279, Confirmation No. 5785, filed August 16, 2025, entitled "Systems and Methods Using Density Field Dynamics for Gravitational and Signal Correction," and U.S. Provisional Patent Application No. 63/865,283, Confirmation No. 5905, filed August 16, 2025, entitled "Systems and Methods Using Density Field Dynamics for Energy Transmission," both of which are incorporated herein by reference.

Field of the Invention

The present invention relates to information science, computing, and communications. More specifically, it relates to systems and methods for encoding, transmitting, and processing information using Density Field Dynamics (DFD) and variations in the one-way speed of light (the "c-field") as a novel physical parameter space.

Background of the Invention

Classical computing relies on charge, spin, or quantum superposition as physical carriers of information. Current paradigms assume the invariance of the one-way speed of light, which restricts electromagnetic (EM) propagation modes to transverse fields. The discovery of measurable variations in one-way c—the c-field—enables longitudinal EM modes and density-dependent signal pathways.

Prior inventions (63/865,279 and 63/865,283) demonstrate the application of Density Field Dynamics (DFD) to gravitational correction and energy transmission. The present disclosure extends these principles to computing and information theory, enabling encoding of information directly into variations of the local c-field propagation parameters.

Summary of the Invention

The invention provides systems and methods for:

1. Encoding information into variations of the c-field.
2. Transmitting information via longitudinal EM wave modes supported by DFD.
3. Processing information by exploiting interference and density-dependent phase shifts in the c-field.
4. Implementing computing architectures, including classical, hybrid, and quantum-like models, that use c-field dynamics as carriers of logic states and entangled correlations.

Detailed Description

Information Encoding

Information is encoded by modulating one-way c magnitudes within a density field. Such modulations may act as discrete symbols (bits, qubits, or higher-dimensional states).

Information Transmission

The modulated c-field supports longitudinal EM propagation that bypasses many limitations of transverse EM (attenuation, line-of-sight, shielding). Transmission fidelity can be enhanced by exploiting density-induced stability modes.

Computing Architectures

A computing unit comprises:

- A DFD-modulator (for imposing information-bearing variations in c).
- A propagation channel (longitudinal EM mode carrier).
- A DFD-detector (for recovering information states).

Logical operations arise from superposition and interference in density-dependent pathways.

Applications

Applications include:

- Ultra-secure communication by encoding into locally measurable c-field values.
- High-density computing architectures beyond silicon and conventional qubits.
- Hybrid systems for AI, cryptography, and sensor networks.

Figure

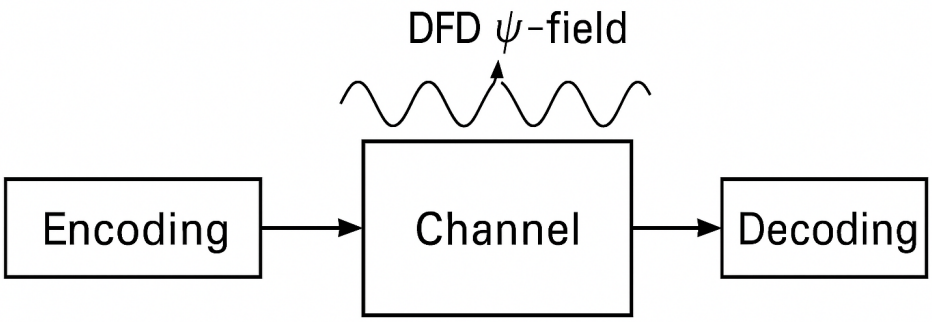

FIG. 1

DFD-modulated communication system

Figure 1: Master schematic of Density Field Dynamics (DFD) for information encoding and computing applications. Encoding module (A) modulates the c-field; propagation through density-modified space (B) supports longitudinal EM transmission; detection and computation unit (C) recovers and processes information states.

Claims

What is claimed is:

1. A system for encoding information into variations of the one-way speed of light (c-field) using Density Field Dynamics.

2. The system of claim 1 wherein said encoded information propagates via longitudinal EM wave modes.

3. The system of claim 1 further comprising detection units that decode information by measuring local variations in c.

4. The system of claim 1 wherein said information encoding supports classical bits, qubits, or higher-dimensional logic states.

5. A computing architecture that performs logic operations using interference of c-field modulations within a Density Field Dynamics framework.

6. A method for secure communication comprising encoding information into locally varying c, transmitting via longitudinal EM waves, and decoding at a receiver.